ENDTERM – QUESTION 14

Lecture 5 – curve fitting

1 By the method of least squares, find the straight line that best fits the following data:

x:	1	2	3	4	5
y:	14	27	40	55	68

2 A simply supported beam carries a concentrated load P(lb) at its midpoint. Corresponding to various values of P, the maximum deflection Y (in) is measured. The data are given below:

P:	100	120	140	160	180	200
Y:	0.45	0.55	0.60	0.70	0.80	0.85

Find a law of the form $Y = a + bP$.

3 By the method of least squares, fit a parabola of the form $y = a + bx + cx^2$, to the following data:

x:	2	4	6	8	10
y:	6.07	12.85	31.47	57.38	91.29

4 Fit a parabola $y = a + bx + cx^2$ to the following data:

x:	1	2	3	4	5	6	7	8	9
y:	2	6	7	8	10	11	11	10	9

5 Fit a least square *geometric curve* $y = ax^b$ to the following data:

x:	1	2	3	4	5
y:	0.5	2	4.5	8	12.5

6 Predict y at $x = 3.75$, by fitting a *power curve* $y = ax^b$ to the given data:

x:	1	2	3	4	5	6
y:	2.98	4.26	5.21	6.10	6.80	7.50

7 Fit the *exponential curve* $y = ae^{bx}$ to the following data:

x:	2	4	6	8
y:	25	38	56	84

8 Fit the curve of the form $y = ae^{bx}$ to the following data:

x:	77	100	185	239	285
y:	2.4	3.4	7.0	11.1	19.6

Lecture 6 – Finite Difference

1 Find the missing terms in the following table:

x:	1	1.5	2	2.5	3	3.5
y:	6	?	10	20	?	1.5

2 Estimate the missing term in the following table:

x:	0	1	2	3	4
$f(x)$	1	3	9	–	81

Lecture 7 – Interpolation

1 Using Newton 's forward formula, fin d the value of $f(1.6)$, if

x:	1	1.4	1.8	2.2
$f(x)$:	3.49	4.82	5.96	6.5

2 From the following table find y when $x = 1.85$ an d 2.4 by Newton's interpolation formula:

x:	1.7	1.8	1.9	2.0	2.1	2.2	2.3
$y = e^x$:	5.474	6.050	6.686	7.389	8.166	9.025	9.974

3 Apply Newton's backward difference formula to the data below, to obtain a polynomial of degree 4 in x:

x:	1	2	3	4	5
y:	1	– 1	1	– 1	1

4 Use Lagrange's interpolation formula to find the value of y when $x = 10$, if the following values of x and y are given:

x:	5	6	9	11
y:	12	13	14	16

5 The following are the measurements T made on a curve recorded by oscilograph representing a change of current I due to a change in the conditions of an electric current.

T:	1.2	2.0	2.5	3.0
I:	1.36	0.58	0.34	0.20

Using Lagrange's formula, find I and $T = 1.6$.

6 Using Lagrange's interpolation, calculate the profit in the year 2000 from the following data:

Year:	1997	1999	2001	2002
Profit in Lakhs of Rs:	43	65	159	248

7 Obtain the Newton 's divided difference interpolating polynomial and hence find $f(6)$:

x:	3	7	9	10
$f(x)$:	160	120	72	63

8 For the following table, find $f(x)$ a s a polynomial in x using Newton's divided difference formula:

x:	5	6	9	11
$f(x)$:	12	13	14	16

9 Find the cubic splines for the following table of values:

x:	1	2	3
y:	– 6	– 1	16

10 The following values of x and y are given:

x:	1	2	3	4
y:	1	5	11	8

Usin g cubic spline, show that $y(1.5) = 2.575$

Lecture 8 – Numerical Differentiation

1 Find the first and second derivatives of the function tabulated below, at the point $x = 1.1$:

x:	1.0	1.2	1.4	1.6	1.8	2.0
$f(x)$:	0	0.128	0.544	1.296	2.432	4.00

2 Given the following table of values of x and y

x:	1.00	1.05	1.10	1.15	1.20	1.25	1.30
y:	1.000	1.025	1.049	1.072	1.095	1.118	1.140

find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at $(a)\ x = 1.05.$ $(b)\ x = 1.25$ $(c)\ x = 1.15.$

3 From the following table, find the values of dy/dx and d^2y/dx^2 at $x = 2.03$.

x:	1.96	1.98	2.00	2.02	2.04
y:	0.7825	0.7739	0.7651 0.	7563	0.7473

4 The following table gives the velocity v of a particle at time t. Find its acceleration at $t = 2$.

t:	0	2	4	6	8	10	12
v:	4	6	16	34	60	94	131

5 Find $f'(6)$ from the following data:

x:	0	2	3	4	7	8
$f(x)$:	4	26	58	112	466	922

6 Using the following data, find x for which y is minimum and find this value of y.

x:	0.60	0.65	0.70	0.75
y:	0.6221	0.6155	0.6138	0.6170

7 Find the value of x for which $f(x)$ is maximum, using the table

x:	9	10	11	12	13	14
$f(x)$:	1330	1340	1320	1250	1120	930

Also find the maximum value of $f(x)$.

Lecture 9 – Numerical Integration

1 Evaluate $\int_0^1 \frac{dx}{1+x}$ applying

(*i*) Trapezoidal rule
(*ii*) Simpson's 1/3 rule
(*iii*) Simpson's 3/8 rule.

2 Evaluate $\int_0^1 \frac{dx}{1+x^2}$ using

(*i*) Trapezoidal rule taking $h = 1/4$.
(*ii*) Simpson's 1/3rd rule taking $h = 1/4$.
(*iii*) Simpson's 3/8th rule taking $h = 1/6$.

3 Given that

x:	4.0	4.2	4.4	4.6	4.8	5.0	5.2
$\log x$:	1.3863	1.4351	1.4816	1.5261	1.5686	1.6094	1.6487

evaluate $\int_4^{5.2} \log x dx$ by

(*a*) Trapezoidal rule

(*b*) Simpson's 1/3 rule,

(*c*) Simpson's 3/8 rule,

4 Use Boole's five point formula to compute $\int_0^{\pi/2} \sqrt{(\sin x)} / dx$

Lecture 10 – Numerical Solution of ODE

1 Employ Picard 's method to obtain, correct to four places of decimals the, solution of the differential equation $dy/dx = x^2 + y^2$ for $x = 0.4$, given that $y = 0$ when $x = 0$.

2 Find an approximate value of y when $x = 0.1$, if $dy/dx = x - y^2$ and $y = 1$ at $x = 0$, using

(*a*) Picard's method (*b*) Taylor's series.

3 Using Taylor's series method, compute $y(0.2)$ to three places of decimals from $\frac{dy}{dx} = 1 - 2xy$ given that $y(0) = 0$.

4 Apply Euler's method to solve $y' = x + y$, $y(0) = 0$, choosing the step length = 0.2. (Carry out six steps).

5 Using Euler's method, find the approximate value of y when $x = 0.6$ of $dy/dx = 1 - 2xy$, given that $y = 0$ when $x = 0$ (take $h = 0.2$).

6 Given $y' = x + \sin y$, $y(0) = 1$. Compute $y(0.2)$ and $y(0.4)$ with $h = 0.2$ using Euler's modified method.

7 Use Runge's method to approximate y when $x = 1.1$, given that $y = 1.2$ when $x = 1$ and $dy/dx = 3x + y^2$.

8 Using the Runge-Kutta method of order 4, compute $y(0.2)$ and $y(0.4)$ from 10 $\frac{dy}{dx} = x^2 + y^2$ $y(0) = 1$, taking $h = 0.1$.

9 Given $dy/dx = x^3 + y$, $y(0) = 2$. Compute $y(0.2)$, $y(0.4)$, and $y(0.6)$ by the Runge-Kutta method of fourth order.

10 Find $y(0.1)$ and $y(0.2)$ using the Runge-Kutta fourth order formula, given that $y' = x^2 - y$ and $y(0) = 1$.